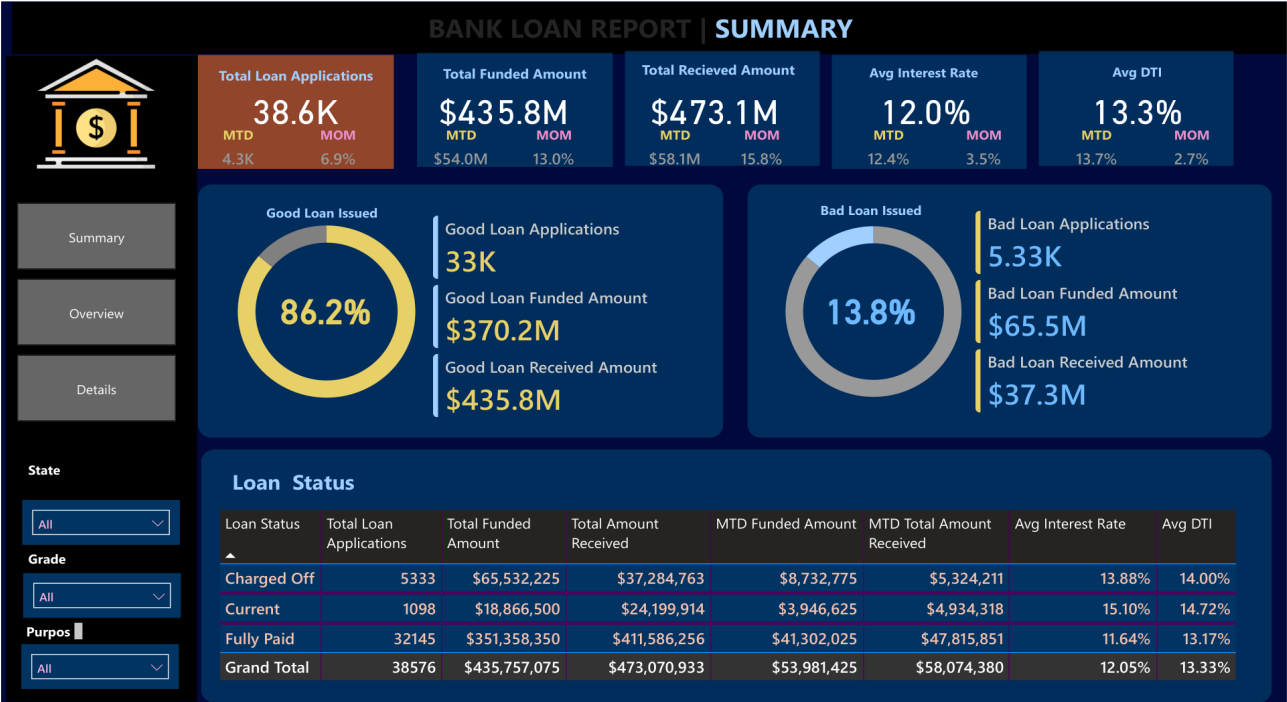


■ Data Analytics Case Study: Bank Loan Risk & Performance

An end-to-end project using SQL Server and Power BI to monitor loan performance, assess portfolio quality, and generate executive business insights.

Created By: **Bijal Bhaliya**

SQL Developer | Power BI Analyst | Data Analyst



Project Objectives

The Business Problem:

The bank required an automated reporting system to monitor lending activity, evaluate portfolio performance, and identify financial risks using data-driven insights.

Project Goals:

- Distinguish between Good Loans and Bad Loans
- Monitor Month-to-Date (MTD) and Month-over-Month (MoM) growth
- Analyze lending trends across states and borrower segments
- Evaluate borrower profiles using DTI, Interest Rate, Employment Length, and Home Ownership

Tools & Technologies Used:

- SQL Server
- Power BI Desktop
- DAX Measures
- Data Visualization & KPI Reporting

The SQL Engine — KPI Engineering

```
SELECT loan_status,  
       COUNT(id) AS Total_Applications,  
       SUM(loan_amount) AS Total_Funded_Amount  
FROM bank_loan_data  
GROUP BY loan_status;
```

Technical Logic:

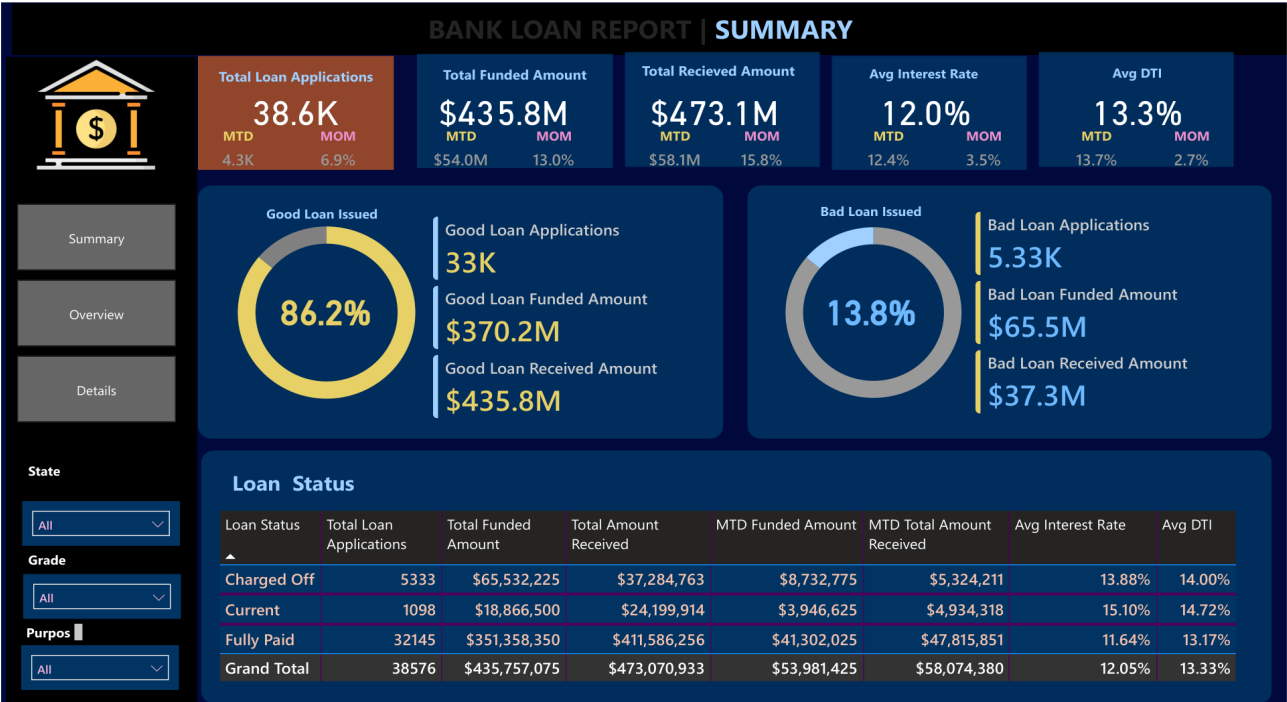
Developed custom SQL queries to validate KPI accuracy and classify loan portfolio quality.

- Fully Paid + Current → Good Loans (86.2%)
- Charged Off → Bad Loans (13.8%)

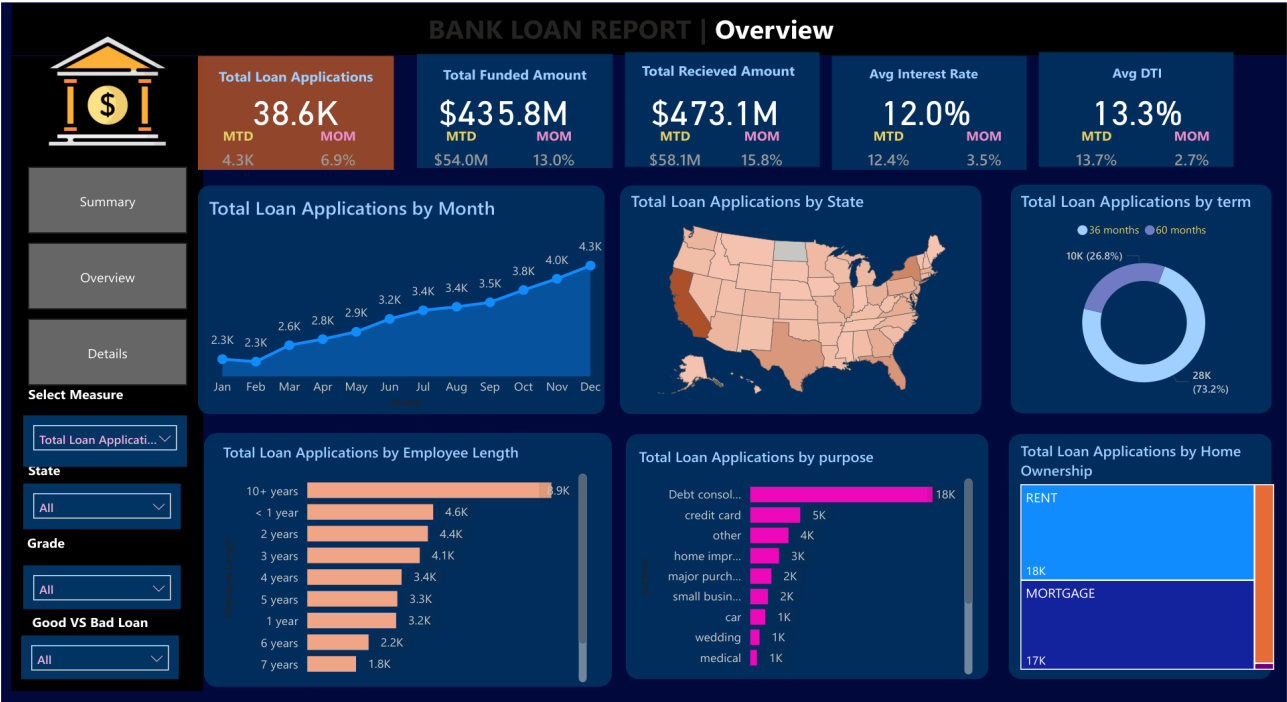
Created KPI calculations for:

- Total Loan Applications
- Funded Amount
- Total Amount Received
- Average Interest Rate
- Debt-to-Income Ratio (DTI)

Dashboard 1 — Executive Summary



Dashboard 2 — Strategic Overview



Advanced Visual Analytics:

- Monthly trend analysis identified seasonal loan patterns
- Regional map analysis highlighted high-performing states
- Donut and Tree Maps revealed borrower segmentation patterns
- Loan Purpose analysis identified Debt Consolidation as the top category
- Employment Length analysis evaluated borrower stability

Dashboard 3 — Operational Details



Operational Intelligence Dashboard:

The Details Dashboard provides a drill-down interface for operational reporting and borrower-level analysis.

Features Included:

- Borrower Profiles
- Loan Amount Tracking
- Employment Length Insights
- Home Ownership Analysis
- Interest Rate Monitoring
- Individual Loan Performance Records

Results & Professional Impact

Project Outcomes:

- Automated reporting for 15+ banking KPIs
- Delivered visibility into \$65.5M of Bad Loan exposure
- Improved monitoring of portfolio performance and repayment trends
- Built executive dashboards for strategic decision-making

Key Skills Demonstrated:

- SQL Query Optimization
- Power BI Dashboard Development
- Financial KPI Reporting
- Data Visualization
- Business Intelligence Analysis